

**Project 2 for Computational Astrophysics (AS8003), Spring 2026 :
Nucleosynthesis networks.**

Total possible points: 22 (2/3 of module grade, 22% of total course grade). Both the written report and the oral presentation (10 minutes + questions/discussion) combine for scoring.

Hand-in deadline: Wednesday April 29 2026, 09.00. Submit through Athena both written report and code.

Late hand-in gives -4p per new 24h delay period.

Your task is to build a nuclear reaction network in Fortran, using the techniques and data sources we have studied in the module.

As first step, choose a type of nucleosynthesis from

1. Big Bang
2. Hydrostatic stellar burning; H, He, C, Ne, O or Si.
3. r-process in a neutron star merger.
4. s-process in an AGB star.
5. Nucleosynthesis in novae.
6. Explosive supernova burning; O or Si.

Your network does not have to be 'realistic' in the sense that all relevant reactions and nuclei are included (this would take you many months to do) - rather, the idea is that you study how composition evolves given a selection of reactions and nuclei. You probably want to keep your network limited to say 5-6 nuclei.

There are a large number of aspects of nuclear burning one can dive deeper into. These include

1. *Analytic versus numeric Jacobian* computation - pros and cons.
2. *Adaptive step-size algorithm.* How should the computer program select which time-step to use next?
3. Solving the linear equation system (in the Newton-Raphson step) by *matrix inversion* vs *iteration*
4. Co-solving the network equations with a *temperature equation*.
5. Investigation of *updated/improved reaction rates* in the literature. Are *screening* and/or *stellar enhancement* factors relevant for your case?

6. Investigate the *degree of stiffness* of your network is, and discuss/explore algorithms from this.
7. Investigate *more advanced implicit algorithms* than backward Euler for optimal performance (while this is often unconditionally stable, it is only first order accurate), e.g. Kaps-Rentrop method (Numerical Recipes, page 730) and Bader-Deuffhard method (Numerical Recipes page 735). Can the sparseness of the Jacobian be utilized?
8. Role of *photodestruction* and *neutrino interactions*.

Choose two aspects from this list that you think sounds particularly interesting (and have some relevance for your case), and in your report explore these two aspects.

Your report is probably around 8 pages including text, a few figures, equations, and tables. Key components are probably (this is just a suggestion of structure, you are free to choose your own)

- A general introduction of the particular type of nucleosynthesis you will study. Some literature suggestions are provided on Athena.
- An outline of which nuclei and reactions you have chosen to include, and why. It can be instructive to plot rate coefficients versus temperature.
- A selection of algorithm with motivation, and a derivation of the discretized equations. An explanation of how your code solves these. You may include a section demonstrating convergence/reliability of the algorithm (this may e.g. include comparison to theoretically expected time-scales, or whether more Newton-Raphson steps improves accuracy).
- A results section, with some explanatory text of how the abundances evolve, and why.
- A section addressing your special topic 1.
- A section addressing your special topic 2.
- A discussion of how your model may be improved physically (which further nuclei and reactions may be important?) and numerically (can your algorithm be further improved for stability/accuracy/speed?).

Good luck and have fun!
Anders